👁️ GlaucoScan

Overview

This project is a deep learning-based desktop application designed to assist in the preliminary inspection of **Fundus Camera Images** for signs of **Glaucoma**.

It utilizes a three-stage computer vision pipeline:

1. **Preprocessing:** This included background removal to eliminate black borders, contrast enhancement using gamma correction and other image augmentation.
2. **UNet Segmentation:** Accurately localizes and segments the optic disc and cup region.
3. **EfficientNet-B3 Classification:** Classifies the cropped region to distinguish between **Non-Glaucoma** and potential **Glaucoma** cases.

The system is built using **PyTorch** and deployed with a user-friendly **PyQt5 GUI**, allowing researchers and medical professionals to perform local, real-time image analysis.

🚀 Features

- **Two-Stage Preprocessing:** Implements robust preprocessing and UNet segmentation to ensure the classification model focuses only on the critical optic disc region, maximizing diagnostic accuracy.
- **High-Performance Backbone:** Utilizes a fine-tuned **EfficientNet-B3** network for fast and reliable feature extraction and classification.
- **Fundus Image Optimized:** The segmentation and preprocessing steps are specifically tailored for varying qualities and sizes of fundus camera images, effectively removing dark backgrounds and centering on the region of interest.
- **Confidence Metrics:** Displays precise probability scores for both the “Glaucoma” and “Non-Glaucoma” categories.
- **Clean Desktop UI:** A fully featured, self-contained desktop interface built with PyQt5, optimizing ease of use for local image analysis.

🛠️ Installation and Setup

Here we provide two ways: python scripts and software

For python scripts

Prerequisites

- Python 3.8+
- NVIDIA GPU (Recommended for faster inference, otherwise runs on CPU)

Environment Setup

1. **Clone the repository:**

```
git clone https://github.com/Brokengm/BIA_group_work.git
cd final_version
```

2. **Create and activate the environment (recommend conda,if you don’t have conda, you can search installation guidance online):**

```
# Using conda
conda create -n glaucoma python=3.9
conda activate glaucoma
```

3. **Install dependencies:**

```
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118 # Use appropriate CUDA version
pip install -r requirements.txt
# The required packages include: PyQt5, numpy, Pillow, timm, scikit-image
```

4. **Files you need:**
For the application to run, you must download the these files and place them directly into the same directory:

   - `384_unet.pth` UNet Segmentation Model
   - `efficientnet.pth` EfficientNet Classification Model
   - `app_gui.py` Main application, GUI (PyQt5) logic, and prediction pipeline entry point.
   - `model.py` Contains UNet and CombinedModel (EfficientNet) class definitions and model loading logic.
   - `preprocess.py` Contains all image processing functions (cropping, CLAHE, segmentation, etc.).
   - `welcome.jpeg` The picture of main page.

If your system is MacOS, you can try our software

Just download GlaucoScan.zip, and decompress.

💡 Usage

Running the Application

1. Ensure you have completed the **Installation and Setup** steps.

2. Run the main application script:

```
python app_gui.py
```

Analysis Steps

1. **Welcome Screen:** Click **“Let’s get started”**.
2. **Load Image:** Click the **“Add image”** button to select a fundus image (`.jpg`, `.png`, etc.). The image will appear in the left panel.
3. **Analyze:** Click **“Run analysis”**. The system will perform:
   - Preprocessing (border removal, contrast enhancement).
   - Optic Disc Segmentation (using UNet).
   - Optic Disc Cropping (centering on the segmented region).
   - Glaucoma Classification (using EfficientNet-B3).
4. **View Results:** The classification result, including the predicted class (`Glaucoma` or `Non-Glaucoma`) and the precise probability scores, will be displayed in the right panel.

⚠️ Disclaimer

This software is intended for course work and educational demonstration only. It must NOT be used for real medical diagnosis or self-assessment of glaucoma. Always consult a qualified medical professional for health concerns.